Project Interim Report
On

Library Management System
(Using Core Java & OOP Concepts)

RU-100-01-00012
Object Oriented Programming with Java

SESSION: 2025-26

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BHILAI, CG

(April 2026)

Abstract

This project designs and implements a Library Management System using Object-Oriented Programming (OOP) in Java. It manages book records by storing details like title, author, and availability, and allows users to search, issue, and return books.
The system reduces manual work and errors by providing an automated solution. It uses classes and objects to represent books and updates their availability dynamically. This project demonstrates key OOP concepts such as encapsulation, constructors, and arrays of objects.

Introduction

A Library Management System is a software application designed to manage library resources efficiently. In traditional libraries, manual record-keeping often leads to errors, misplacement, and inefficiency. This project uses Java and Object-Oriented Programming (OOP) principles to develop a simple and effective system that stores book details such as title, author, and availability. It allows users to perform key operations like searching for books, issuing and returning them, and viewing all available books. The system also provides a menu-driven interface for easy user interaction, making library management more organized and efficient.

Objectives

- To manage book records efficiently within the library
- To keep track of issued and available books accurately
- To reduce manual work and minimize human errors
- To automate basic library operations like search, issue, and return
- To apply Object-Oriented Programming (OOP) concepts
- To understand the use of classes, objects, and constructors
- To implement arrays of objects for managing multiple books
- To develop a simple, structured, and user-friendly system

Scope

This project is suitable for small libraries such as school or college libraries, where basic book management is required. It provides an efficient way to store book details, track availability, and manage issuing and returning of books, reducing manual work and errors. In the future, the system can be enhanced by adding database support for permanent storage, a graphical user interface for better usability, and user authentication to improve security and make it suitable for larger libraries.

System Requirements

Category	Requirement
OS	Windows / Linux / macOS
RAM	4 GB minimum
Java	JDK 8 or above
IDE	VS Code / Eclipse / IntelliJ / Edit Plus
Build	javac or IDE built-in compiler

Methodology

Thesystemfollows a 7-step execution workflow:

Step 1 — Input: User selects operation (search, issue, return, display) using menu-driven input.

Step 2 — Object Creation: Library object is created containing an array of Book objects with title, author, and availability.

Step 3 — Search Operation: System compares input title with stored books and displays result if found.

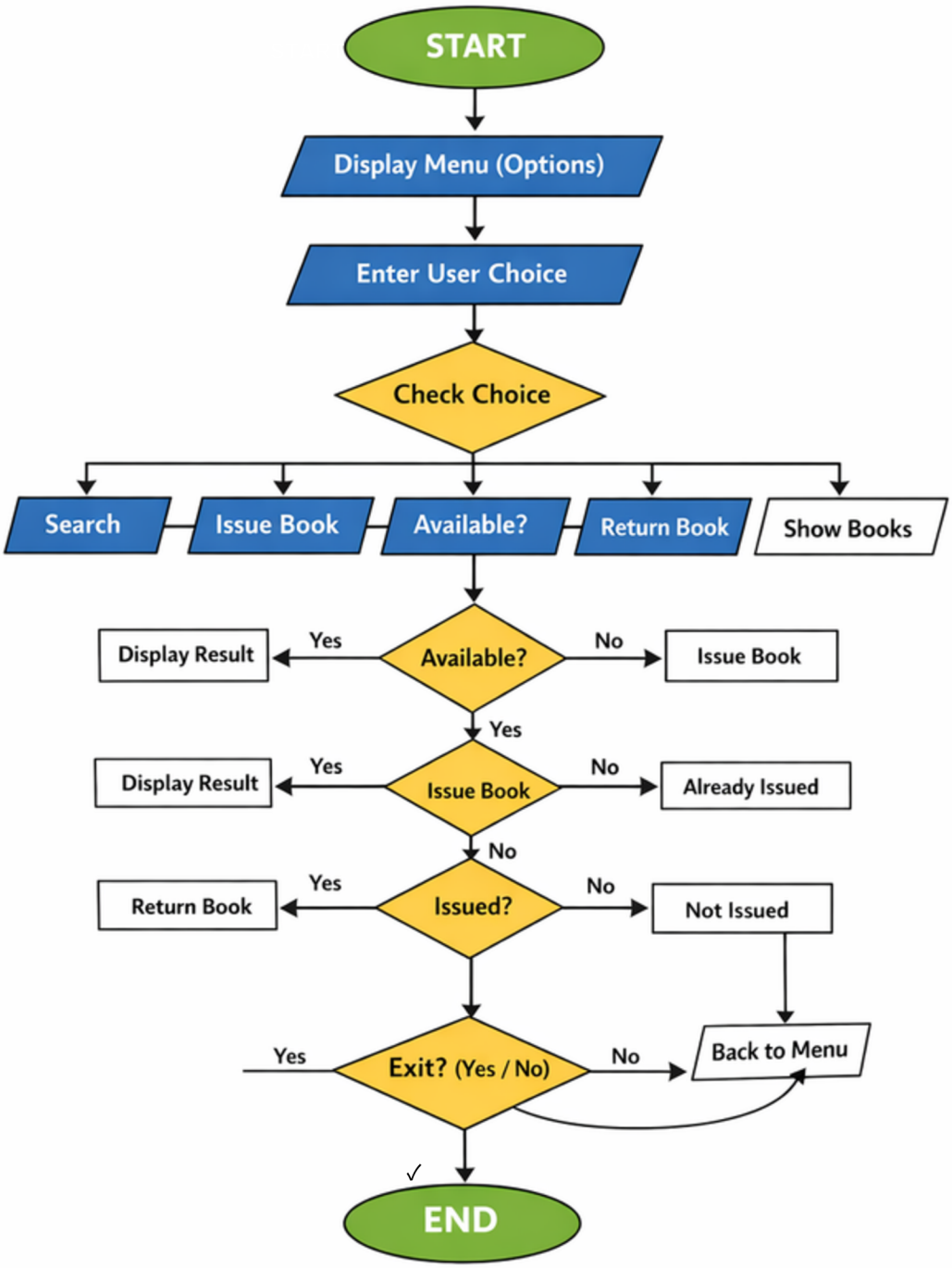
Step 4 — Issue Book: If book is available, set isAvailable = false; otherwise show already issued message.

Step 5 — Return Book: If book is issued, set isAvailable = true; otherwise show not issued message.

Step 6 — Display Books: All books are displayed with their details and availability status.

Step 7 — Result: System repeats operations until user exits, updating book status dynamically.

Flowchart


START
Display Menu (Options)
Enter User Choice
Check Choice
Search
Issue Book
Available?
Return Book
Show Books
Display Result
Yes
Available?
No
Issue Book
Yes
Display Result
Yes
Issue Book
No
Already Issued
No
Return Book
Yes
Issued?
No
Not Issued
Yes
Exit? (Yes / No)
No
Back to Menu
END

Implementation

Book Class

Stores book details using private variables, initializes them using a constructor, and provides methods to issue, return, and manage availability.

Library Class

Manages an array of Book objects, performs search using loops, updates book status during issue/return, and displays all books.

Main Class

Provides a menu-driven interface, takes user input using Scanner, and calls library methods.

Sample Input / Output

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                                    Code

cd "/Users/harshitkumar/Desktop/PROJECT JAVA/" && javac project4.java && java project4
harshitkumar@Harshits-MacBook-Air PROJECT JAVA % cd "/Users/harshitkumar/Desktop/PROJECT JAVA/" && javac project4.java && java project4

📚 LIBRARY MANAGEMENT SYSTEM
1. Show Books
2. Search Book
3. Issue Book
4. Return Book
5. Exit
Enter your choice: 1

Available Books:
1. Java Basics by James Gosling - Available
2. Data Structures by Mark Weiss - Available
3. Python Fundamentals by Guido van Rossum - Available

📚 LIBRARY MANAGEMENT SYSTEM
1. Show Books
2. Search Book
3. Issue Book
4. Return Book
5. Exit
Enter your choice: 2
Enter book title to search: Java Basics
Book Found:
Java Basics by James Gosling - Available

📚 LIBRARY MANAGEMENT SYSTEM
1. Show Books
2. Search Book
3. Issue Book
4. Return Book
5. Exit
Enter your choice: 3
Enter book title to issue: Java Basics
Book issued successfully

📚 LIBRARY MANAGEMENT SYSTEM
1. Show Books
2. Search Book
3. Issue Book
4. Return Book
5. Exit
Enter your choice: 4
Enter book title to return: Java Basics
Book returned successfully

📚 LIBRARY MANAGEMENT SYSTEM
1. Show Books
2. Search Book
3. Issue Book
4. Return Book
5. Exit
Enter your choice: 5
Exiting...
harshitkumar@Harshits-MacBook-Air PROJECT JAVA %
```

Future Enhancements

- Add database support (MySQL) for permanent storage
- Develop GUI using Swing or JavaFX
- Implement user login and authentication system
- Add fine calculation for late returns
- Introduce unique Book ID system
- Enable adding and removing books dynamically
- Implement advanced search (by title, author, or ID)
- Maintain transaction history (issue/return records)
- Add notification system for due dates
- Support multiple users (admin and user roles)

Gantt Chart

S. No	Task	Start	End
1	Requirement Analysis & Planning	10 April	10 April
2	System Design (Classes & Flow)	10 April	11 April
3	Implement Book Class	11 April	11 April
4	Implement Library Class	11 April	12 April
5	Develop Main Program	12 April	12 April
6	Testing & Debugging ✗	12 April	13 April
7	Documentation	13 April	13 April
8	Final Review & Submission	13 April	13 April

Conclusion

The Library Management System successfully demonstrates the use of Object-Oriented Programming concepts in Java. It provides a simple and efficient way to manage library operations such as searching, issuing, and returning books. The project improves understanding of real-world applications of OOP and enhances programming skills

References

[Book:

[1] H. Schildt, Java: The Complete Reference, McGraw-Hill

Website:

[2] Oracle, "Java Documentation,"

https://docs.oracle.com